

BS Computer Science — UAF

Complete 1st–8th Semester Syllabus Overview

University: University of Agriculture Faisalabad (UAF)

Program: Bachelor of Science in Computer Science (BSCS)

Total: 150 Credit Hours | 8 Semesters

This short guide summarizes the official UAF BS(CS) Scheme of Studies. It is designed as a quick reference rather than a detailed course syllabus.

Degree at a Glance

Area	What you study
Programming	Programming fundamentals, OOP, advanced OOP, modern languages, visual programming
Core CS	Data structures, algorithms, OS, architecture, networks, databases, compilers
Software	Software Engineering I & II, Web Programming, Software Project
AI / Advanced	Artificial Intelligence, Automata, System Programming, Distributed Databases
Math / Support	Calculus, linear algebra, differential equations, statistics, electronics
Final Stage	AI + 10-credit Software Project in Semester 8

Semester-wise Syllabus — Page 2

Semester 1 — 16 Credit Hours

Code	Course	Cr. Hrs
CS-301	Introduction to Computers	3
CS-303	Introduction to Computer Programming	4
PHY-305	Basic Electronics	3
Math-303	Calculus and Analytical Geometry	4
Eng-301	Introductory Exercises in Reading, Comprehension and Communication Skills	2

Semester 2 — 16 Credit Hours

Code	Course	Cr. Hrs
CS-302	Fundamentals of Algorithms	3
CS-304	Discrete Structures	4
Math-304	Multi Variable Calculus	4
Eng-302	Advanced Exercises in Reading, Comprehension and Communication Skills	2
MBA-306	Financial Accounting	3

Semester 3 — 18 Credit Hours

Code	Course	Cr. Hrs
CS-401	Object Oriented Programming	3
CS-403	Data Base Systems	3
Math-405	Differential Equations	4
PHY-405	Circuit Theory	3
MBA-407	Human Resource Management	3
IS-401 / SSH-301(A)	Islamic Studies / Ethics	2

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Semester 4 — 22 Credit Hours

Code	Course	Cr. Hrs
CS-402	Data Structures	3
CS-404	Digital Logic Design	3
CS-406	Software Engineering - I	4
CS-408	Modern Programming Languages	4
MBA-404	Financial Management	3
Math-406	Linear Algebra	3
SSH-402	Pakistan Studies	2

Semester 5 — 22 Credit Hours

Code	Course	Cr. Hrs
CS-501	Software Engineering II	4
CS-503	Web Programming	4
CS-505	Analysis of Algorithms	3
CS-507	Computer Organization & Assembly	4
CS-509	Numerical Analysis	3
Stat-507	Statistics and Probability	4

Semester 6 — 22 Credit Hours

Code	Course	Cr. Hrs
CS-502	Operating System Concepts	4
CS-504	Computer Architecture	4
CS-506	Data Communication	3
CS-508	Computer Graphics	4
CS-510	Advanced Object Oriented Programming	4
CS-512	Automata Theory	3

Semester-wise Syllabus — Page 4

Semester 7 — 20 Credit Hours

Code	Course	Cr. Hrs
CS-601	System Programming	4
CS-603	Compiler Construction	4
CS-605	Visual Programming	4
CS-607	Computer Networks	4
CS-609	Distributed Database Systems	4

Semester 8 — 14 Credit Hours

Code	Course	Cr. Hrs
CS-602	Artificial Intelligence	4
CS-604	Software Project	10

What This Degree Builds

- **Programming foundation:** You start with programming, then move into OOP, advanced OOP and multiple programming paradigms.
- **Problem solving:** Algorithms, data structures, analysis of algorithms and discrete structures develop computational thinking.
- **Systems:** Operating systems, computer architecture, assembly, system programming and compiler construction explain how software runs on computers.
- **Data & networking:** Database systems, distributed databases, data communication and computer networks cover data storage and communication.
- **Software development:** Software Engineering I/II, Web Programming, Visual Programming and the final Software Project cover development practices.
- **AI:** Artificial Intelligence appears in Semester 8, giving the degree a direct foundation for later AI/ML specialization.
- **Mathematics:** Calculus, multivariable calculus, differential equations, linear algebra and statistics support computing and analytical work.

Possible Career / Specialization Directions

Direction	Relevant BSCS subjects
AI / Machine Learning	Artificial Intelligence, Algorithms, Statistics, Linear Algebra, Programming
Software Development	OOP, Data Structures, Software Engineering, Web Programming, Visual Programming
Data / Databases	Database Systems, Distributed Database Systems, Algorithms, Statistics
Cyber / Systems	Operating Systems, Networks, Computer Architecture, System Programming
Web Development	Programming, OOP, Web Programming, Database Systems
Further Study	MS Computer Science, AI, Data Science, Software Engineering and related computing fields (subject to admission requirements)

Important: UAF's LMS currently lists additional/current BSCS courses such as Machine Learning, Big Data Analytics, Cloud Computing, Digital Image Processing, Mobile Application Development and Internet of Things. These may belong to a newer or different scheme/course offering, so students should verify their own batch's official scheme before treating them as required courses.

Source: University of Agriculture Faisalabad — official BS (Computer Science) Scheme of Studies. Prepared as a concise study/reference guide.